

☒ EASY

☐ ADVANCED MATH

Advanced Math

10 questions · ~15 min

10

SAT QUESTION BANK

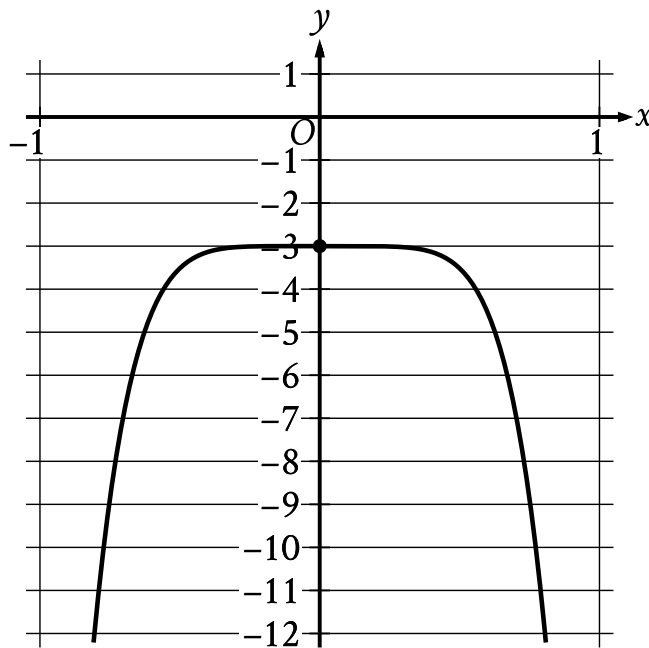
2026-05-29

Q1

ADVANCED MATH

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

- A. $4ts16s^2 - 14ts$
- B. $4ts16t^2s^2 - 14t$
- C. $4t^2s16ts - 14s$
- D. $4t^2s16s^2 - 14t$



- In quadrant 3, the curve rises to its maximum at point (0 comma negative 3).
- In quadrant 4, the curve then falls.

The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is 0, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6

7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

$$x = 3$$

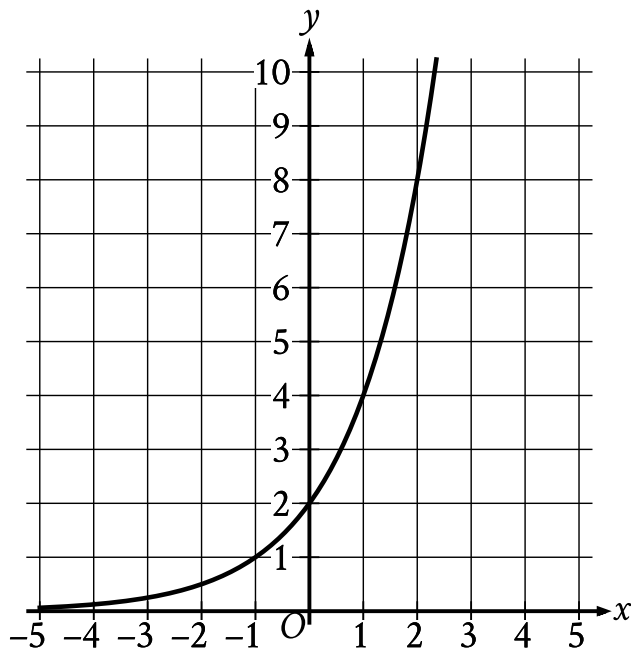
$$y = 15 - x^2$$

A solution to the given system of equations is x, y . What is the value of xy ?

- A. 432
- B. 54
- C. 45
- D. 18

Which expression is equivalent to $12x + 27$?

- A. $129x + 1$
- B. $2712x + 1$
- C. $34x + 9$
- D. $39x + 24$



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 2).
 - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
 - (negative 1 comma 1)
 - (0 comma 2)
 - (2 comma 8)

What is the y-intercept of the graph shown?

- A. 0,0
- B. 0,2
- C. 2,0

D. 2, 2

Q6

ADVANCED MATH

$$c - 7 = 25p + k$$

The given equation relates the positive numbers c , p , and k . Which equation correctly expresses c in terms of p and k ?

A. $c = 25p + k + 7$

B. $c = 25p + k - 7$

C. $c = 725p + k$

D. $c = \frac{25p + k}{7}$

Q7

ADVANCED MATH

Which expression is equivalent to $8yzy7z$?

A. $56y^2z^2$

B. $56y^2z$

C. $56yz$

D. $16yz$

Q8

ADVANCED MATH

The function g is defined by $gx = \sqrt{8x+1}$. What is the value of g^3 ?

- A. $\frac{5}{8}$
- B. $\frac{25}{8}$
- C. 5
- D. 25

Q9

ADVANCED MATH

$$x = 8$$

$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point x, y in the xy -plane. What is the value of y ?

- A. 8
- B. 24
- C. 64
- D. 72

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

- A. $23xx^2 + 2x + 9$
- B. $9x23x^3 + 2x^2 + 1$
- C. $x23x^2 + 2x + 9$
- D. $34x^3 + x^2 + x$